SMaHT Sample & File Nomenclature

Resource for SMaHT Network Members
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Description

The names of SMaHT samples and files, along with their accession identifiers, are the primary identifiers that are permanent and immutable. The SMaHT sample names are for the biosamples distributed by the Tissue Procurement Center (TPC) to the Genome Characterization Centers (GCCs) and the Tools and Technology Development groups (TTDs) for sequencing. The SMaHT file names are for the data generated from and analyzed for the SMaHT samples by the Data Analysis Center (DAC) of the SMaHT Network (“Network”), which are released on the SMaHT Data Portal (https://data.smaht.org/).

Overview

The names of the SMaHT samples and files are unique identifiers that are permanent and immutable, and also contain metadata codes, which were included to make “the data about the data” (i.e., metadata) more accessible and obvious to downstream users.

This document describes the naming schema and metadata codes in the SMaHT sample and file names. Each metadata code in the sample and file names is delimited by a hyphen (“-”). As placeholders, “`#`” in this document indicates a single-digit integer (and similarly, “`##`” for double-digit numbers, “`###`” for three-digit numbers, etc.), and “`A`” indicates an alphabetical letter.

Complete SMaHT Naming Schema
[Project][Donor ID] - [Tissue ID] - [Tissue aliquot & Tissue core IDs] - [Sex & Age] - [Sequencing platform & Sequencing assay code] - [Center code] - [Unique Accession ID] - [Analysis info] . [File format]

Donor

Donor Identifiers

The donors from whom tissue samples were analyzed for the SMaHT benchmark and production studies were procured by the Tissue Procurement Center (TPC) at NDRI. The donor identifiers are assigned by the TPC, following the schema below.

Donor Naming Schema: ProjectDonorID
Benchmark donors: **ST**###
Production donors: **SMHT**###
Examples:
ST001
SMHT001

- **Project**: **ST** is designated for a benchmark donor, and **SMHT** for a production donor.
    - Note: In rare cases, some tissues from the TPC were sent to TTDs to test their protocols without generating data for the benchmark or production studies in SMaHT. For these non-benchmarking, non-production donors, "**SN**" is used.
- **Donor ID**: Associated with a unique donor. The Donor ID refers to a TPC's donor kit ID, which is assigned by the TPC. It is a three-digit pre-generated identifier (derived from a kit used to procure a tissue sample) that references the donor. 001–150 are reserved for production donors and 001–004 for benchmarking donors.

Donor Identifiers for Cell Lines used for the SMaHT Benchmark Study

Some benchmark samples were derived from established cell lines, such as COLO829 and HapMap, not from TPC donors. For these cell line samples, the donor identifiers are as shown in **Table 1**.

Table 1. Donor identifiers for SMaHT benchmark cell lines.

Donor ID	Cell line description

SMHTCOLO829T	COLO829 tumor cell line from ATCC, used in the benchmark studies by the SMaHT Network
SMHTCOLO829BL	COLO829BL normal lymphoblast cell line from ATCC, used in the benchmark studies by the SMaHT Network
SMHTCOLO829BLT50	Admixture of COLO829 and COLO829BL cell lines at a 1:49 mixture ratio, respectively, created by the University of Washington and used in the benchmark studies by the SMaHT Network
SMHTCOLO829TVAI	COLO829 tumor cell line from ATCC, used for STORM-Seq at Washington University in St. Louis
SMHTCOLO829BLVAI	COLO829BL normal lymphoblast cell line from ATCC, used for STORM-Seq at Washington University in St. Louis
SMHTCOLO829BLTVAI	Admixture of COLO829 and COLO829BL cell lines at a 1:1 mixture ratio, respectively, created by WashU-VAI
SMHTHAPMAP6	Admixture of six HapMap cell lines, created for the Network at Coriell (see the SMaHT Data Portal for more details)
SMHTLBLA2	LB-LA2 fibroblast cell line created by the Network
SMHTLBIPSC1	iPSC line derived from Clone #1 from the parental LB-LA2 fibroblast cell line, created by the Network
SMHTLBIPSC2	iPSC line derived from Clone #2 from the parental LB-LA2 fibroblast cell line, created by the Network
SMHTLBIPSC4	iPSC line derived from Clone #4 from the parental LB-LA2 fibroblast cell line, created by the Network
SMHTLBIPSC52	iPSC line derived from Clone #52 from the parental LB-LA2 fibroblast cell line, created by the Network
SMHTLBIPSC60	iPSC line derived from Clone #60 from the parental LB-LA2 fibroblast cell line, created by the Network

* LB-LA2 fibroblast and the iPSC cell lines are described in Fasching L *et al.* (2021) Science.

Tissue Sample

Tissue Identifiers

The tissue identifiers are assigned by the TPC. The TPC provides the metadata for tissue samples to GCCs and TTDs, as well as to the DAC.

Tissue Naming Schema: ProjectDonorID – TissueID – TissueAliquotIDCoreID
Benchmark tissues: **ST### – #A – ###A#** **Production tissues**: **SMHT### – #A – ###A#**

Note:

1. Unless otherwise noted in the metadata, tissues from all benchmark donors are homogenized tissues (to remove spatial/cellular heterogeneity in the samples distributed across multiple GCCs for sequencing). At the specific requests of GCCs or TTDs, non-homogenized, intact benchmark donor tissues were distributed for sequencing.
2. Tissues from all production donors are intact tissues. They are *not* homogenized.
3. For intact tissues, solid tissues are referred to as "tissue aliquots", which subsequently get sectioned into multiple "tissue cores" or labeled as a "tissue specimen" without further sectioning. This depends on the size of the available tissue material (tissue size = Aliquot > Specimen > Core).
4. Tissue sample identifiers from the TPC are unique and immutable. GCCs and TTDs provide the DAC with the metadata for the samples they receive from the TPC without any alteration.

- **Tissue ID**: A tissue identifier automatically generated by the TPC's Rhythm system, corresponding to the TPC's tissue protocol ID. This code consists of a single number ("1" or "3"), followed by an alphabetical designation (A-ZZ), which corresponds to a specific combination of tissue type and tissue preservation method (see **Tables 2A and 2B**).
    - For data generated from combining multiple tissue samples (e.g., mutations identified across multiple tissue types), the **Tissue ID** in the file name is denoted as **MT**, to indicate multiple tissues. E.g., **SMHT001 – MT**
- **Tissue Aliquot IDs**: Generated by the TPC. A tissue aliquot refers to either a solid tissue (e.g., liver) or a liquid tissue aliquoted in a vial (e.g., whole blood). A tissue aliquot ID is a three-digit number, e.g., 001.
    - For intact tissues (in either a tissue core or tissue specimen form):
        - Tissue aliquot refers to spatially numbered solid tissue aliquots from longitudinally bisected or non-bisected tissue samples (**Figure 1**). Serial cores are subsampled from within a tissue aliquot. Their identifiers are designated by a three-digit number, which corresponds to a specific location in the tissue aliquot, as shown in **Figures 1 and 2**.
        - E.g., SMHT001 – 3S – **002**
    - For homogenized tissues used for the SMaHT benchmark studies or non-solid production tissues:
        - File names for the data derived from these samples will have the value "**X**" for the **Tissue Aliquot ID** to indicate that the notion of a tissue core does not apply to homogenized or non-solid tissues.
        - E.g., ST001 – 1Q – **X**
- **Tissue Core IDs**: Provided for the tissue core samples from the TPC. Composed of a letter (between **A–F**) and a digit (between **1–6**), which is associated with a location on a 6-by-6 grid (**Figure 2**), indicating the specific spatial position of the core sample in a tissue aliquot. See **Figure 2** for more details.
    - For a tissue specimen sample that is neither sectioned nor cored, the letters between **S–W** followed by numbers **1–9** were reserved for **Tissue Core IDs**.

These identifiers are not associated with a location on the grid and thus have no associated spatial information.

- For the homogenate benchmark tissues and non-solid production tissues: **Tissue Core ID** is "**X**", to indicate that the notion of a core is not applicable for these tissue types.
  - Example for benchmark tissue homogenates: **ST001 – 1Q – XX**
  - Example for non-solid production tissues: **SMHT001 – 3A – XX**

- For the file names for data generated from combining multiple tissue aliquots and/or multiple cores, **Tissue Aliquot ID** and **Core ID** will be "**MA**" (to indicate multiple aliquots) and "**MC**" (to indicate multiple cores), respectively.
  - Example for multiple cores from the same aliquot: SMHT001 – 3S – **002MC**
  - Example for multiple cores from multiple aliquots: SMHT001 – 3S – **MAMC**

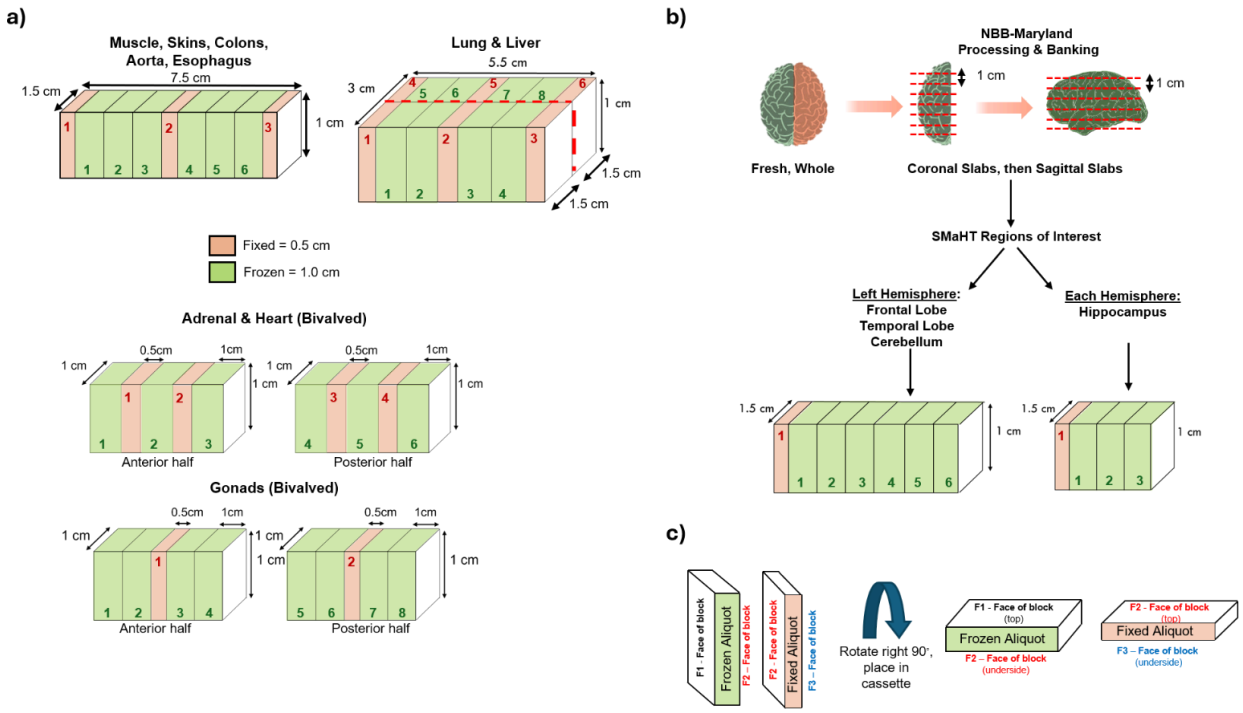


Figure 1. SMaHT Tissue Recovery Schema. (a) Schema of the spatial relationship between fixed (pink) and frozen (green) tissue aliquots recovered for each type of non-brain tissue for the SMaHT project. Solid tissue samples from muscle, skins (suprapubic abdomen (non-exposed) and calf (sun-exposed)), colons (ascending and descending), aorta, and esophagus are recovered and sectioned. Larger tissue samples from the lung and liver are recovered and sectioned into medial and lateral halves, as denoted by the dotted red line in the upper right panel. Small organs (adrenal glands and gonads) and the heart ventricle are bisected into anterior and posterior halves before aliquoting. Please note that this is a representative diagram and that the size of each tissue aliquot can vary greatly. **(b)** Recovery and processing schema of the brain for SMaHT. **(c)** Standardized orientation of both fixed and frozen aliquots

prior to preservation from solid tissues. Frozen aliquots from mucosal tissues (skin, colon, aorta, and esophagus) are all placed serosa side down in cassettes for processing of a full-thickness sample (all cell layers).

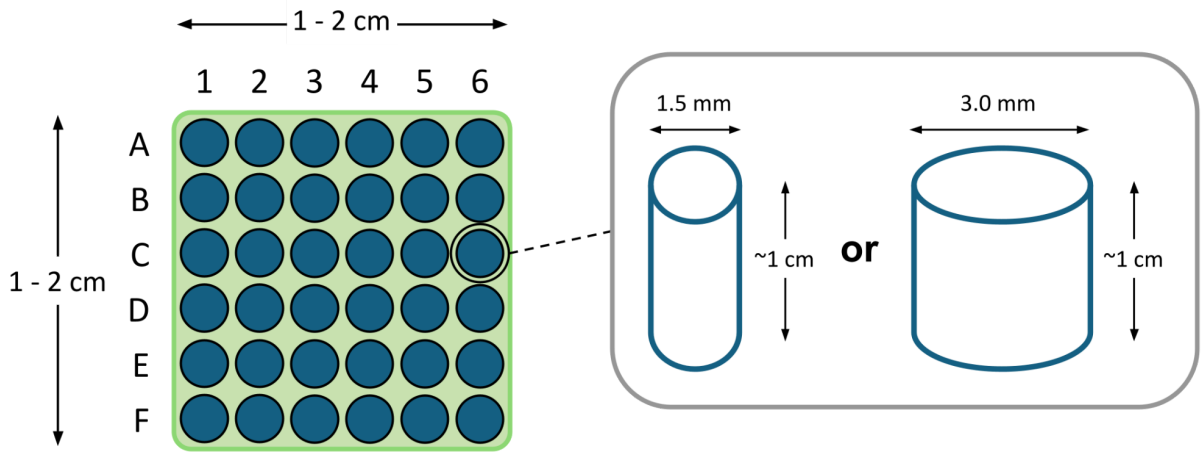


Figure 2. Standardized coring schema for frozen solid tissue aliquots. Cores are subsampled from a tissue aliquot (typically 1-2 cm per side) following a standardized grid of 6x6 positions, and each core is assigned a core ID. The core ID consists of a letter between A and F to denote the vertical position of the core, followed by a single numerical digit between 1 and 6 to denote the horizontal position of the core within the aliquot. "**X**" represents a null value, indicating samples that do not get sub-sampled (e.g., non-solid tissues). Cores are either 1.5 mm or 3.0 mm in diameter and are approximately 1 cm in depth. Please note that this is a **representative** diagram – the size of each aliquot and the number and arrangement of cores taken from each aliquot vary from sample to sample.

Table 2A. Benchmark tissue IDs, which contain information for both the tissue types and tissue preservation methods. Snap-frozen tissue samples are sent to GCCs or TTDs for sequencing, while fixed tissue samples are sent to pathologists for their review.

Tissue ID	Tissue Name	Preservation Method	Notes
1A	Liver	Snap Frozen	For both homogenate and non-homogenate samples used in the SMaHT benchmark studies
1B	*unassigned*	*N/A*	
1C	Liver	Fixed	
1D	Lung	Snap Frozen	For both homogenate and non-homogenate samples used in the SMaHT benchmark studies

1E	*unassigned*	*N/A*	
1F	Lung	Fixed	
1G	Colon	Snap Frozen	For both homogenate and non-homogenate samples used in the SMaHT benchmark studies
1H	*unassigned*	*N/A*	
1I	Colon	Fixed	
1J	Skin	Snap Frozen	Tissue specimen (~10 cm)
1K	Skin	Snap Frozen	Larger tissue cores, each ~1 cm in diameter, were made from intact skin tissues
1L	Skin	Fixed	
1M/N/O/P	*unassigned*	*N/A*	
1Q	Brain, Frontal lobe	Snap Frozen	For both homogenate and non-homogenate samples used in the SMaHT benchmark studies

Table 2B. Production tissue IDs, which contain information for both the tissue types and tissue preservation methods. Snap-frozen tissue samples are sent to GCCs or TTDs for sequencing, while fixed tissue samples are sent to pathologists for their review.

Tissue ID	Tissue Name	Preservation method
3A	Blood, Whole*	Snap Frozen
3B	Buccal swab	Fresh
3C	Esophagus	Snap Frozen
3D		Fixed
3E	Colon, Ascending	Snap Frozen
3F		Fixed
3G	Colon, Descending	Snap Frozen
3H		Fixed
3I	Liver	Snap Frozen
3J		Fixed

3K	Adrenal gland, Left	Snap Frozen
3L		Fixed
3M	Adrenal gland, Right	Snap Frozen
3N		Fixed
3O	Aorta, Abdominal	Snap Frozen
3P		Fixed
3Q	Lung	Snap Frozen
3R		Fixed
3S	Heart, Left ventricle	Snap Frozen
3T		Fixed
3U	Testis, Left	Snap Frozen
3V		Fixed
3W	Testis, Right	Snap Frozen
3X		Fixed
3Y	Ovary, Left	Snap Frozen
3Z		Fixed
3AA	Ovary, Right	Snap Frozen
3AB		Fixed
3AC	Dermal fibroblast*	Cultured Cells
3AD	Skin, Calf	Snap Frozen
3AE		Fixed
3AF	Skin, Abdomen	Snap Frozen
3AG		Fixed
3AH	Muscle	Snap Frozen
3AI		Fixed
3AJ	Brain	Fresh
3AK	Frontal lobe, Brain, Left hemisphere	Snap Frozen
3AL	Temporal lobe, Brain, Left hemisphere	Snap Frozen

3AM	Cerebellum, Brain, Left hemisphere	Snap Frozen
3AN	Hippocampus, Brain, Left hemisphere	Snap Frozen
3AO	Hippocampus, Brain, Right hemisphere	Snap Frozen
3AP	Frontal lobe, Brain, Left hemisphere	Fixed
3AQ	Temporal lobe, Brain, Left hemisphere	Fixed
3AR	Cerebellum, Brain, Left hemisphere	Fixed
3AS	Hippocampus, Brain, Left hemisphere	Fixed
3AT	Hippocampus, Brain, Right hemisphere	Fixed

* Whole blood was collected from either the clavicle or femoral vein from a *post-mortem* donor. A total of 12 mL of blood is collected at one time, pooled and mixed with anticoagulant. The pooled mixture is then aliquoted into twelve 1 mL aliquots and flash-frozen for distribution to sequence at GCCs/TTDs.
* Fibroblasts are isolated from a patch of fresh calf skin and are not derived from a single cell.

Cell Line

The cell-line samples used for the SMaHT benchmark experiments were prepared by a GCC or TTD, not distributed by the TPC. As such, the notion of TPC protocols and intact tissue aliquots does not apply to these samples. Thus, **Tissue IDs** and **Tissue Aliquot IDs** are “X” in the file names to indicate a null value, e.g., SMHTCOLO829T – **X** – **X**.

Sample- and Assay-Related Metadata Codes in the File Name

The file name contains additional metadata codes delimited by hyphens (“-”) as shown below in bold.

[Project][Donor ID] - [Tissue ID] - [Tissue aliquot & Tissue core IDs] - **[Sex & Age]** - **[Sequencing platform & Sequencing assay code]** - **[Center code]** - [Unique Accession ID] - [Analysis info] . [File format]

- **Sex & Age**: Sex and age of the donor, i.e., “M” for male and “F” for female, followed by the donor age in years. “N” will be used for sex and/or age when they are unknown or not applicable.
    - E.g., the HapMap mixture samples that combined cell lines from multiple individuals: SMHTHAPMAP6 – X – X – **NN**

- **Sequencing platform & Sequencing assay code**: The sequencing platform code consists of an alphabetical letter (**Table 3A**) and is followed by a three-digit number that corresponds to an experimental assay type (**Table 3B**).
    - For data generated using multiple sequencing platforms and/or assays (e.g., variants identified using Illumina and PacBio data), both codes are denoted as "X" in the file name. E.g., `SMHTCOLO829BLT50 - X - X - M45 - XX`
- **Center code**: Corresponds to a data-generating center that submitted original data to the DAC (e.g., raw sequencing data) (**Table 4**).

Table 3A. Sequencing platform codes.

SMaHT code	Sequencing platform
A	Illumina NovaSeq X, Illumina NovaSeq X Plus
B	PacBio Revio HiFi
C	Illumina NovaSeq 6000
D	ONT PromethION 24
E	ONT PromethION 2 Solo
F	ONT MinION Mk1B
G	Illumina HiSeq X
H (Note: Deprecated)	*Illumina NovaSeq X Plus*
I	BGI DNBSEQ-G400
J	Element AVITI
K	Illumina NextSeq 2000
L	PacBio Sequel IIe
M	Ultima Genomics UG 100
N	PacBio Onso
O	MGI DNBSEQ-T7
P	Roche Axelios 1

Table 3B. Experimental assay codes.

SMaHT Code	Assay Name	Description
000		(Null or not-applicable)

[001-100: DNA-based assays]		
001	WGS	DNA, PCR-free, Bulk, Whole genome sequencing (WGS)
002	PCR WGS	DNA, PCR, Bulk, WGS
003	Ultra-Long WGS	DNA, PCR-free, Bulk, Ultra-Long WGS
004	Fiber-seq	DNA, PCR-free, Bulk, Fiber-seq
005	Hi-C	DNA, Bulk, Hi-C
006	Bulk NTSeq	DNA, Bulk, NTSeq
007	CODEC	DNA, Bulk, Duplex-seq, CODEC
008	Bot-seq	DNA, Bulk, Duplex-seq, Bot-seq
009	NanoSeq	DNA, Bulk, Duplex-seq, NanoSeq
010	scNanoSeq	DNA, Single-cell, Duplex-seq, scNanoSeq
011	DLP+	DNA, Single-cell, DLP+
012	Microbulk MALBAC WGS	DNA, Microbulk, MALBAC-amplified WGS
013	Single-cell MALBAC WGS	DNA, Single-cell, MALBAC-amplified WGS
014	Microbulk PTA WGS	DNA, Microbulk, PTA-amplified WGS
015	Single-cell PTA WGS	DNA, Single-cell, PTA-amplified WGS
016	scDip-C	DNA, Single-cell, scDip-C
017	CompDuplex-seq	DNA, Bulk, Duplex-seq, CompDuplex-seq
018	scCompDuplex-seq	DNA, Single-cell, Duplex-seq, scCompDuplex-seq
019	Strand-seq	DNA, Bulk, Strand-seq
020	scStrand-seq	DNA, Single-cell, scStrand-seq
021	HiDEF-seq	DNA, Bulk, Duplex-seq, HiDEF-seq
022	HAT-seq	DNA, Bulk, HAT-seq
023	Microbulk HAT-seq	DNA, Microbulk, PTA-amplified, HAT-seq
024	scHAT-seq	DNA, Single-cell, PTA-amplified, HAT-seq
025	META-VISTA-seq	DNA, Bulk, Duplex-seq, META-VISTA-seq

026	Microbulk META-VISTA-seq	DNA, Microbulk, Duplex-seq, META-VISTA-seq
027	scMETA-VISTA-seq	DNA, Single-cell, Duplex-seq, META-VISTA-seq
028	TEnCATS	DNA, Bulk, TEnCATS
029	L1-ONT	DNA, Bulk, L1-ONT
030	ppmSeq	DNA, Bulk, Duplex-seq, ppmSeq
031	SBX-D	DNA, Bulk, Duplex-seq, SBX-D
[101-200: RNA-based assays]		
101	RNA-seq	RNA, Bulk, RNA-seq
102	Kinnex	RNA, Bulk, Kinnex
103	snRNA-seq	RNA, Single-cell, snRNA-seq
104	STORM-seq	RNA, Single-cell, STORM-seq
105	Tranquil-seq	RNA, Single-cell, Tranquil-seq
[201-300: Chromatin-based or other epigenome-profiling assays]		
201	ATAC-seq	Chromatin, Bulk, ATAC-seq
202	CUT&Tag	Chromatin, Bulk, CUT&Tag
203	varCUT&Tag	Chromatin, Bulk, varCUT&Tag
204	sc-varCUT&Tag	Chromatin, Single-cell, sc-varCUT&Tag

Table 4. Codes for centers in the SMaHT Network. GCCs, TTDs, DAC, and TPC, as well as the contact PI of each center, are shown below.

Center Code in the Data Portal	**Center Category**	**Full Name of the Center**	**Contact PI**
bcm	GCC	Baylor College of Medicine	Richard Gibbs
broad	GCC	Broad Institute of MIT and Harvard	Kristin Ardlie

nygc	GCC	New York Genome Center	Nicolas Robine
uwsc	GCC	University of Washington and Seattle Children's Research Institute	Jimmy Bennett
washu	GCC	Washington University in St. Louis and Van Andel Institute	Ting Wang
bcm1	TTD	Baylor College of Medicine	Chuck Zong
bcm2	TTD	Baylor College of Medicine	Fritz Sedlazeck
bch1	TTD	Boston Children's Hospital	Christopher Walsh
bch2	TTD	Boston Children's Hospital	Sangita Choudhury
broad1	TTD	Broad Institute of MIT and Harvard	Fei Chen
cwru	TTD	Case Western Reserve University	Fulai Jin
dfci	TTD	Dana-Farber Cancer Institute	Kathleen Burns
mayo	TTD	Mayo Clinic	Alexej Abyzov
nyu	TTD	New York University	Gilad Evrony
stfd	TTD	Stanford University	Alexander Urban
umass	TTD	University of Massachusetts	Thomas Fazzio
umich	TTD	University of Michigan	Ryan Mills
uutah	TTD	University of Utah	Gabor Marth
wcnygc	TTD	Weill Cornell Medicine & New York Genome Center	Dan Landau
dac	DAC	Harvard Medical School	Peter Park
tpc	TPC	National Disease Research Interchange	Thomas Bell

Data Types, Software, and Other Information in the File Name

The unique accession identifier for a file, as well as other information related to data analysis and types, is included in the file name as indicated below in bold.

[Project][Donor ID] - [Tissue ID] - [Tissue aliquot & Tissue core IDs] - [Sex & Age] - [Sequencing platform & Sequencing assay code] - [Center code] - **[Unique Accession ID]** - **[Analysis info]** . **[File format]**

- **Unique Accession ID**: A unique accession ID for every file is automatically generated at the DAC, with the "SMA" prefix (for SMaHT). This accession ID can be used to search for the file and the associated metadata in the SMaHT Data Portal via a URL, as shown below.
    - `https://data.smaht.org/[accession ID]`
    - E.g., `https://data.smaht.org/SMAAR1LD95K6`
- **Analysis info** and **File format**: Indicates a software tool (and its version) used to analyze the data and generate the file. If multiple tools are used, they are delimited by an underscore ("_").
    - **[Software tool names and versions; mandatory] _ [Reference genome version; optional] _ [Data type; optional] _ [Other analysis info as needed]**
        - **Tool name and version (mandatory in the file name):** Software name and version used to analyze the data, e.g., `sentieon_bwamem_202308.01`.
        - **Genome version (Table 5A):** Reference genome used to align sequences.
            - Example: `SMHT039-3U-006B3-M74-A001-washu-SMAAR1LD95K6-sentieon_bwamem_202308.01_GRCh38.aligned.sorted.cram`
        - **Data type (Table 5B)**
            - Indicates additional information related to the analysis, delimited by an underscore ("_") in cases where multiple analyses are performed on the data.
            - VCF:
                - For VCFs containing a single variant type, that type is included in the file name.
                    - E.g., VCF file containing SNVs from Strelka: `…SMAFIVZQIVDV-strelka2_2.9.10_GRCh38.snv.vcf.gz`
                - For VCFs containing multiple variant types, the variant types are not included in the file name.
                    - E.g., VCF file containing multiple variant types from longcallD: `…SMAFI8PI4XU2-longcalld_v0.0.8_GRCh38.vcf.gz`
            - For donor-specific genome assembly files (e.g., FASTA, chain files), haplotypes, and the direction to convert genomic coordinates from a source genome to a target genome ("Source-To-Target Genome") are included in the file name.
                - E.g., `…SMASFTYEDLTK-minimap2_2.28_paf2chain_0.1.0_DSAToGRCh38.chain`

- For RNA-Seq data, the GENCODE version and data types (e.g., gene and transcript isoform quantification levels) are included in the file name.
    - E.g., …SMAFITE6PD91-rsem_v1.3.3_GRCh38_**gencode_v47_isoform**.tsv

Table 5A. Genome version codes.

Reference Genome	**Code in the file name**
GRCh38 without ALT contigs	GRCh38
GRCh38 with ALT contigs	GRCh38_ALT
T2T CHM13	CHM13
Donor-specific genome assembly	DSA

Table 5B. Data type codes.

Data Type	**Code in the file name**
Single nucleotide variants	snv
Short insertions and deletions	indel
Structural variants, including large insertions and deletions, duplications, inversions, and translocations	sv
Copy number variants	cnv
Mobile element insertions	mei
Reference conversion	[Source genome]To[Target genome]
Donor-specific genome assembly haplotype	hapX, hapY, hapX1, hapX2
Gene expression level	gene
Transcript isoform expression level or other isoform-level information	isoform
Exon/Intron junction annotations	junction
Full-length, non-concatemer (FLNC) Kinnex reads	flnc

Aligned consensus reads from Duplex-seq data	`consensus`
Metadata information	`metadata`